

Standing pushup rep count error and its fix

Before:

The application only checked the elbow angles and sideways orientation of the user meaning if the user stood sideways to the camera and simply moved their arms back and forth like a standing chest press or a wall pushup, the app would count it as a floor pushup.

Cause:

- The elbow angle for a standing arm extension is mathematically identical to a floor pushup (180-90 degrees). So, the app lacked “spatial awareness” to know if the user was actually horizontal on the floor or vertical on their feet.

Fix:

- I implemented a spatial geometry check by comparing the distance between the shoulder and the ankle on the X-axis (Horizontal) vs the Y-axis (Vertical).

In a real push-up, the body is horizontal so the distance between the user’s shoulder and ankle on the X-axis (length) should be much larger than the distance on the Y-axis (height)

So, if the height (Y) is greater than the length (X), the user is standing up so with the implementation of the Spatial Geometry Check the code now kills the logic execution immediately and returns a feedback “Lye down on the floor!”

```
// Horizontal Orientation Check
// Comparing X distance (length) to Y distance (height) between shoulder and ankle
double bodyLengthX = (leftShoulder.x - leftAnkle.x).abs();
double bodyHeightY = (leftShoulder.y - leftAnkle.y).abs();

// If Height (Y) is greater than Length (X), the user is standing up.
if (bodyHeightY > bodyLengthX) {
    feedback = "Lye down on the floor!";
    hasFormError = true;
    return; // Stop processing further logic if the user is standing
}
```

Figure 1: Horizontal Orientation Check

- If $Y > X$: The user is vertical (standing)
- If $X > Y$: The user is horizontal (lying down)